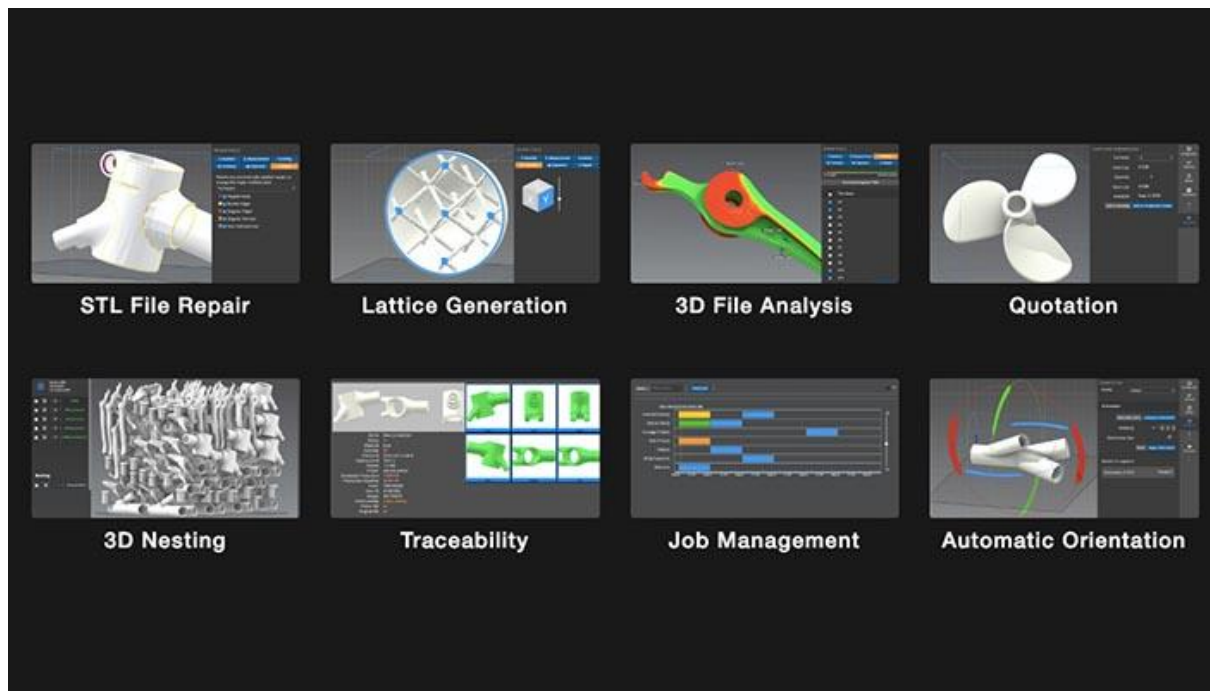




COURSE CONTENT

- 4.1.1 STL files**
- 4.1.2 Overview of RP Software's**
- 4.1.3 Other Popular Software's**

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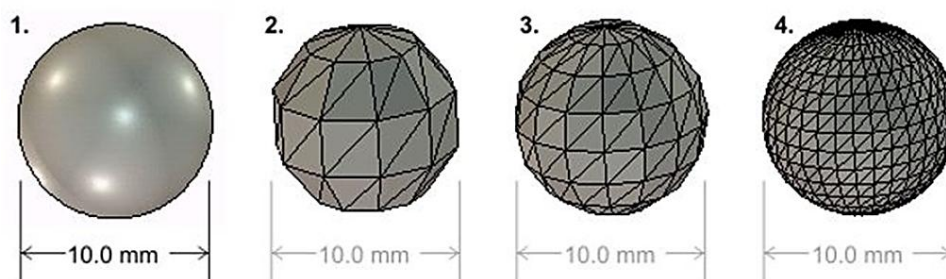
4.1.1 STL FORMAT

STL (Standard Tessellation Language) is a file format native to the STereo Lithography CAD software created by 3D Systems.

- Tessellation is piecewise approximation of surfaces of 3D CAD model using series of triangles.
- The STL file connects the surface of the model in an array of triangles and consists of the X, Y and Z coordinates of the three vertices of each surface triangle, as well as an index that describes the orientation of the surface normal.
- It is widely used for Rapid Prototyping (RP) and Computer-Aided Manufacturing (CAM). This file format is supported by many other software packages.
- STL files describe only the surface geometry of a three-dimensional object without any representation of colour, texture or other common CAD model attributes.
- The STL format specifies both ASCII and binary representations.
- STL files determines the resolution of the print. A model with a high resolution will have a smooth surface with a high level of detail but will take more time to be sliced.
- This format is used as industry standard in Rapid Prototyping.

<pre> solid print facet normal 0.00000e+00 1.00000e+00 0.00000e+00 outer loop vertex 0.00000e+00 0.00000e+00 2.00000e+01 vertex 0.00000e+00 0.00000e+00 0.00000e+00 vertex 1.00000e+01 0.00000e+00 2.00000e+01 endloop endfacet facet normal 0.00000e+00 1.00000e+00 0.00000e+00 outer loop vertex 1.00000e+01 0.00000e+00 2.00000e+01 vertex 0.00000e+00 0.00000e+00 0.00000e+00 vertex 1.00000e+01 0.00000e+00 0.00000e+00 endloop endfacet </pre>	
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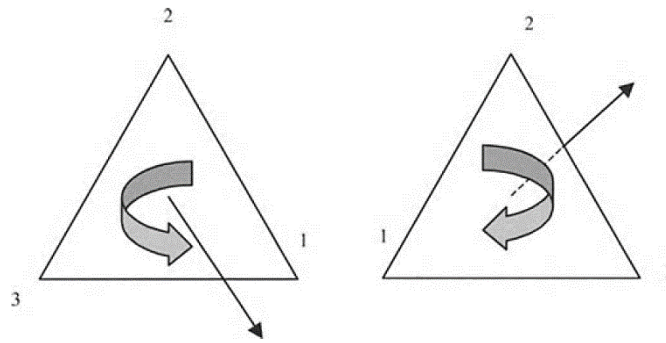
Sample STL file



STL file with different resolutions

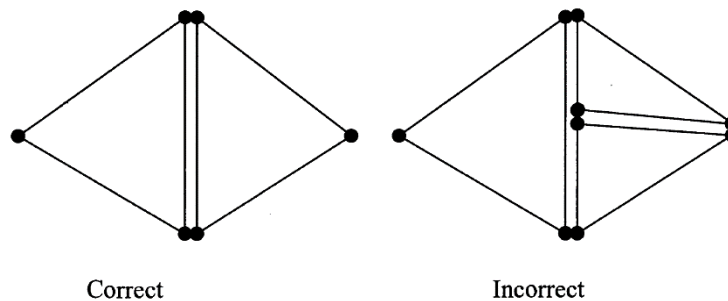
- Two very important requirements must be followed during STL file generation:
 - Data about triangle vertices must be stored in the file in an ordered fashion to identify interior or exterior surfaces. A clockwise vertex ordering defines an

interior surface, and an anti-clockwise ordering the exterior surface. The righthand rule is applied.



Righthand rule

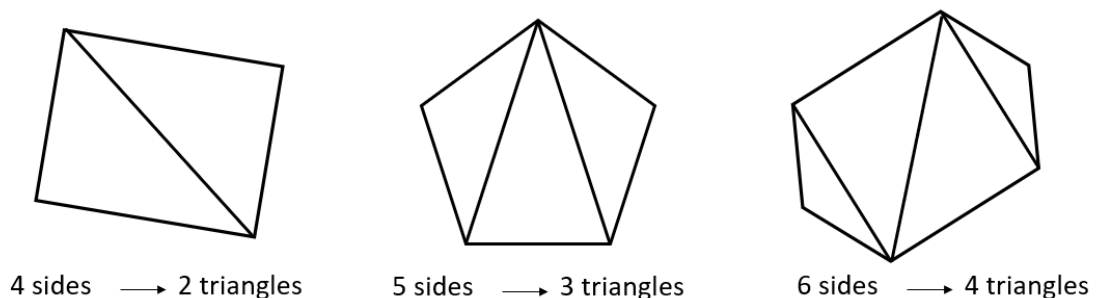
- A triangle must share exactly two common vertices with each adjacent triangle. This is known as the vertex-to-vertex rule.



Vertex-to-vertex rule

PROCESS OF APPROXIMATING SURFACES WITH TRIANGLES:

- Each 3D form is made out of polygons.
- A polygon is defined as a flat shape which is bounded by a closed circuit.
- Each polygon with n sides can be represented using $(n - 2)$ triangles



Why is it used in RP?

- The STL file format has become the standard data transmission format, and is the format required to interact with Quickparts.
- Slicing a part using STL format is easier compared to other formats, such as Boundary representation (B-rep) and Constructive Solid Geometry (CSG), which will need complex geometric reasoning and data conversion.
- Almost all of today's CAD systems are capable of producing an STL file.
- For the user the process is often as simple as selecting File, Save As, and STL.

- In all cases, export your STL file as a Binary file. This saves on time and file size.
- There is absolutely no loss of data whatsoever due to compression and encryption involved in the RP file format.
- If you create a RP file from a STL file and then convert it back to a STL file, you will end up with exactly the same STL file.

4.1.2 OVERVIEW OF RP SOFTWARE'S

The modern engineering ecosystem relies heavily on digital tools that allow designers, engineers, and medical professionals to visualize, modify, analyze and prepare 3D data. Three important tools used widely in engineering and biomedical applications are:

- SolidView – For viewing, measuring, and annotating CAD models
- Materialise Magics – For preparing and repairing 3D files for additive manufacturing
- Materialise Mimics – For converting CT/MRI scans into accurate 3D anatomical models

These tools bridge the gap between design, manufacturing, healthcare, and research.

1. SOLIDVIEW

SolidView (now called Freeform) is a 3D visualization and communication software that allows users especially those without CAD expertise to view, inspect, and interact with CAD models. It removes the dependency on complex CAD software for everyday engineering communication.

Key Features:

a) 3D Model Viewing:

- SolidView can open various CAD file formats such as STL, IGES, STEP, and DXF, enabling compatibility with multiple CAD systems.
- It provides smooth real-time rotation, zooming, and panning, allowing easy examination of complex models.
- The software uses lightweight rendering, making it capable of loading and displaying large assemblies efficiently.

b) Measurement & Inspection:

- The software supports essential measurements such as linear distances, angles, radii, and diameters.
- Users can create section views to analyze internal features of a model.
- Advanced versions allow volume and surface area calculations for quick engineering checks.

c) Markup & Collaboration:

- SolidView allows users to add annotations, labels, comments, and simple dimensions directly on the model.
- These marked views can be exported as images or included in reports for sharing with team members or vendors.
- It is a useful tool during design reviews and communication with stakeholders who do not use CAD software.

d) Model Translation:

- The software can convert CAD files into neutral or lightweight formats for easier sharing.

- This helps protect the original CAD files by distributing simplified versions for viewing only.
- It supports collaboration across teams using different software tools.

Industrial Use Cases

- Manufacturing engineers inspecting designs before fabrication
- Managers reviewing models without CAD training
- Quality inspection teams checking tolerances
- Communication between OEMs and vendors
- Technical illustrators preparing manuals and part catalogs

2. MATERIALISE MAGICS

Materialise Magics is a comprehensive AM data preparation software, considered the industry standard in 3D printing workflow. It provides powerful features to repair, modify, optimize, and prepare digital models before they are sent to a 3D printer.

Key Features:

a) CAD Import & Analysis:

- Magics supports importing a wide range of CAD formats and converts them into mesh data for editing.
- It provides tools to analyze model quality, thickness, and errors before printing.

b) STL Repair & Optimization:

- The software can repair common mesh issues such as holes, bad edges, inverted normals, and intersecting triangles.
- Magics ensures the model becomes watertight and printable.
- Optimization tools help improve print quality by refining the mesh where needed.

c) Build Preparation for 3D Printing:

- Users can orient the model to achieve better strength, surface finish, and reduced support usage.
- Automatic and manual support generation tools help prepare models for metal and polymer 3D printing.
- Nesting tools allow arranging multiple parts efficiently on the build plate, reducing time and material usage.

d) Manufacturing Workflow Tools:

- Magics provides features for production planning, part labeling, job documentation, and quality inspection.
- Lattice and lightweighting tools help reduce weight while maintaining structural integrity.
- It supports integration with various industrial 3D printing systems.

Industrial Applications

- Aerospace and automotive printing
- Medical implants & prosthetics
- Tooling and jigs
- Industrial prototypes
- High-volume AM service bureaus

3. MATERIALISE MIMICS

Materialise Mimics is specialized software used for converting medical imaging data (CT, MRI, micro-CT) into accurate 3D anatomical models. It is widely used in healthcare, biomedical engineering, surgery planning, and research.

Key Features:

a) Medical Image Processing:

- Mimics imports DICOM data from CT, MRI, and other imaging systems.
- It provides tools for segmentation to isolate bones, organs, blood vessels, soft tissues, or implants.
- Users can visualize anatomical structures in 2D slices and 3D models.

b) 3D Model Generation:

- The software converts segmented regions into accurate 3D surface models representing patient-specific anatomy.
- The models can be smoothed, refined, and edited as needed.
- Mimics ensures high accuracy suitable for engineering analysis and medical use.

c) Measurements & Analysis:

- Users can measure distances, angles, thickness, volume, and shape directly on the anatomical model.
- These tools assist in diagnosis, pre-operative planning, and research.

d) Export & Integration:

- Models can be exported in STL format for 3D printing or simulation.
- Mimics integrates with Materialise 3-matic for advanced mesh optimization suitable for FEA or CFD.
- The models can also be exported to other engineering tools for implant design or biomechanical studies.

Applications:

- Preoperative surgical planning
- Designing patient-specific implants and prosthetics
- Medical education and training
- Research in biomechanics
- Production of anatomical models for teaching and diagnosis

4. MAGICS COMMUNICATOR:

Magics Communicator is a lightweight 3D viewer designed to help users view, inspect, and discuss 3D files that have been prepared using Materialise Magics. It allows engineers, clients, and production teams to review models without needing the full Magics software, making it ideal for communication and approval workflows in AM.

Key Features:

a) Viewing STL and CAD Files:

- Magics Communicator can open STL and various CAD files that have already been processed or exported from Magics.

- Users can visually inspect the geometry, check mesh quality, and explore different viewing modes to understand the design clearly.
- The tool is optimized to load files quickly so even large models can be examined without performance issues.

b) Review of Part Orientation, Supports, and Build Strategies:

- The software allows users to see how the part has been oriented for 3D printing in Magics.
- It also displays all support structures, enabling reviewers to verify whether the part is properly supported for successful printing.
- Users can examine build strategies such as layer direction and the positioning on the build platform, helping ensure print reliability.

c) Annotation and Comments:

- Magics Communicator includes simple markup tools that allow users to add comments, highlight areas, and point out issues directly on the model.
- These annotations help in clearly communicating necessary corrections or approvals between engineers and clients.
- The comments can be saved and shared so everyone involved can follow the same feedback trail.

d) Lightweight Collaboration Tool:

- Because Magics Communicator does not require a full Magics license, it is very accessible for external partners, customers, and non-technical stakeholders.
- It ensures that all reviewers see the same 3D preparation details support structures, orientations, and build settings exactly as prepared by the AM engineer.
- This reduces errors and misunderstandings during the manufacturing process.

4.1.3 OTHER POPULAR SOFTWARE'S

Following are some Popular Software Used in 3D Modeling, Additive Manufacturing

3D Modelling Softwares

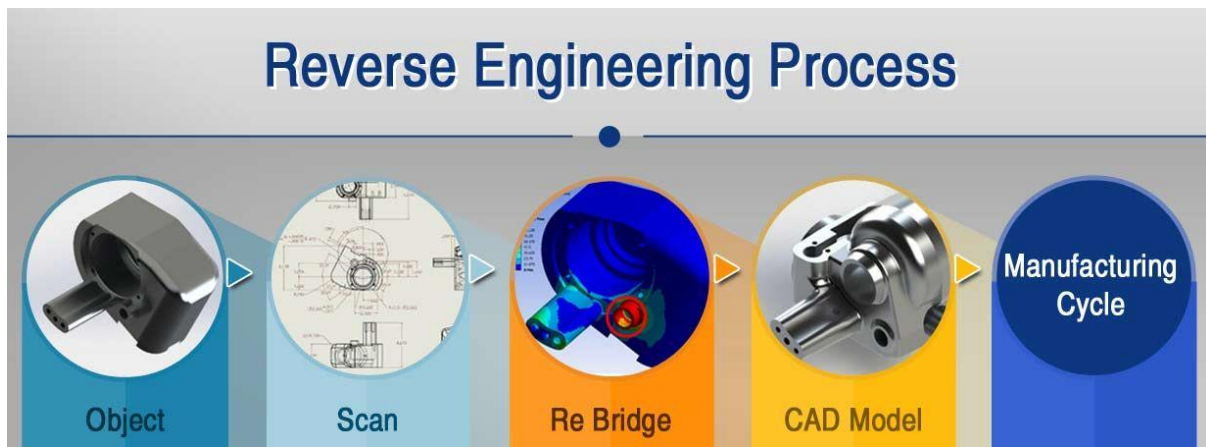
- SolidWorks
- CATIA
- Siemens NX
- PTC Creo
- Autodesk Inventor
- Rhino 3D
- Blender
- FreeCAD

Slicing & Printer Preparation Software

- Ultimaker Cura (offline mode)
- PrusaSlicer
- Lychee Slicer
- Simplify3D
- FlashPrint

Internet-Based AM Softwares

- Materialise CO-AM Platform
- Autodesk Fusion 360
- Onshape
- 3DPrinterOS
- GrabCAD Print
- Ultimaker Digital Factory
- EOSCONNECT
- HP 3D Center
- Formlabs Dashboard
- nTopology
- Siemens Teamcenter (Cloud AM workflows)
- AMFG (Additive MES)
- MakerBot CloudPrint
- Shapr3D Cloud Sync
- Tinkercad



COURSE CONTENT

- 4.2.1 Basic Concept**
- 4.2.2 Steps in Reverse Engineering**
- 4.2.3 Digitization Techniques**
- 4.2.4 Data Processing for Additive Manufacturing**

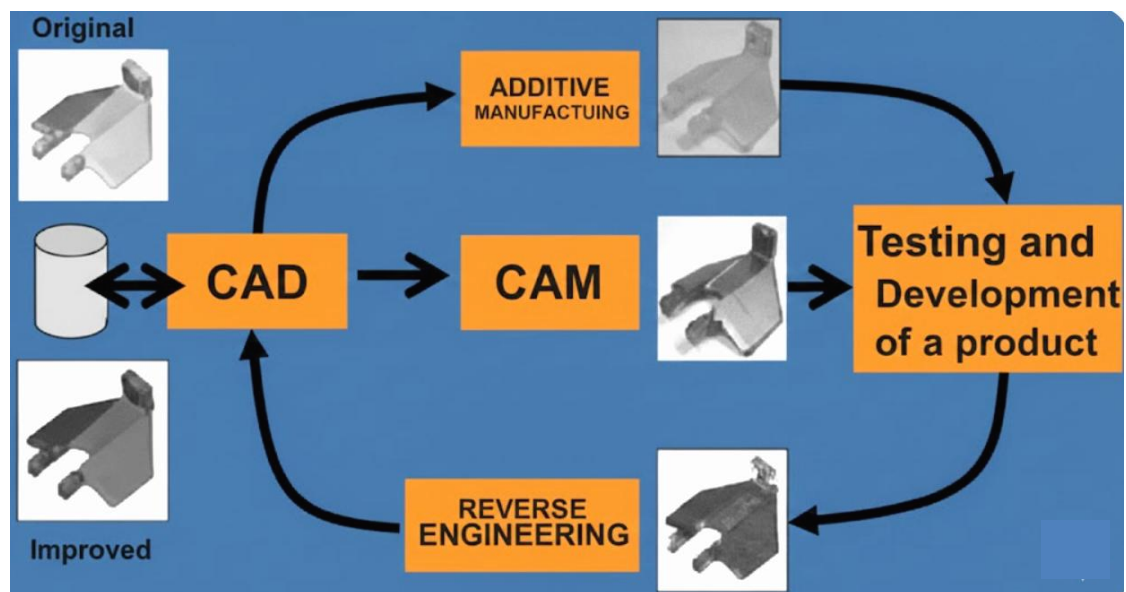
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4.2.1 BASIC CONCEPT

Engineering involves the processes of designing, manufacturing, assembling, and maintaining products and systems. Broadly, engineering can be classified into two categories forward engineering and reverse engineering. Forward engineering is the traditional development pathway in which the designer moves from high-level abstractions and logical concepts to the physical realization of a system.

However, many situations arise where a physical part or product exists without any technical documentation, such as engineering drawings, bills of material, or CAD data. In such cases, the need emerges to recreate or redesign the component based solely on the physical model.



Forward v/s Reverse Engineering

Definition:

The process of duplicating an existing part, subassembly, or product without drawings, documentation, or a computer model is known as Reverse Engineering.

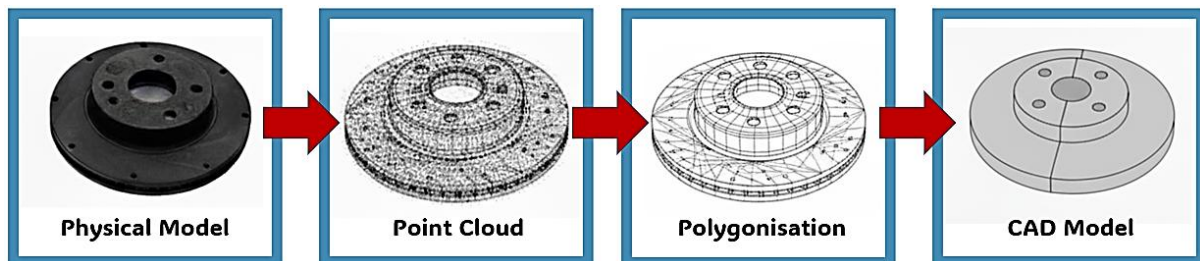
Today, reverse engineering has become an essential tool in various fields such as manufacturing, industrial design, automotive engineering and jewelry design and reproduction. For example, when a new car is introduced into the market, competitors often purchase and dismantle it to understand how it was built and how it functions. In software engineering, reverse engineering is used to study and improve existing source code. In fields like automotive styling, designers often sculpt models using clay, plaster, wood or foam; yet a precise CAD model is required for manufacturing. As product shapes become increasingly complex and organic, creating exact shapes directly in CAD becomes challenging, and there is no guarantee that the CAD model will replicate the sculpted form accurately.

Why Use Reverse Engineering?

Reverse engineering is applied across many industries for a wide range of practical, technical, and strategic reasons. Some of the major reasons for using reverse engineering include:

- The original manufacturer no longer exists, but a customer still needs the product, for e.g., aircraft spare parts required after an aircraft has been in service for several years.
- The original manufacturer of a product no longer produces the product, for e.g., when the product has become obsolete.
- The original product design documentation has been lost or never existed.
- Creating data to refurbish or manufacture a part for which no CAD data exists, or for which existing data has become obsolete or lost.
- Inspection and/or quality control, comparing a fabricated part to its CAD description or to a standard item.
- Eliminating bad features of a product; for example, excessive wear may indicate areas where a product must be improved.
- Strengthening the good features of a product based on long-term usage experience.
- Analyzing the good and bad features of competitors products.
- Exploring new avenues to improve product performance and features.
- Creating 3D data from a model or sculpture for animation in games and movies.
- Creating 3D data from an individual, model, or sculpture to create, scale or reproduce artwork.
- Architectural and construction documentation and measurement.
- Fitting clothing or footwear to individuals and determining the anthropometry of a population.
- Generating data to create dental or surgical prosthetics, tissue-engineered body parts or for surgical planning.
- Documentation and reproduction of crime scenes.

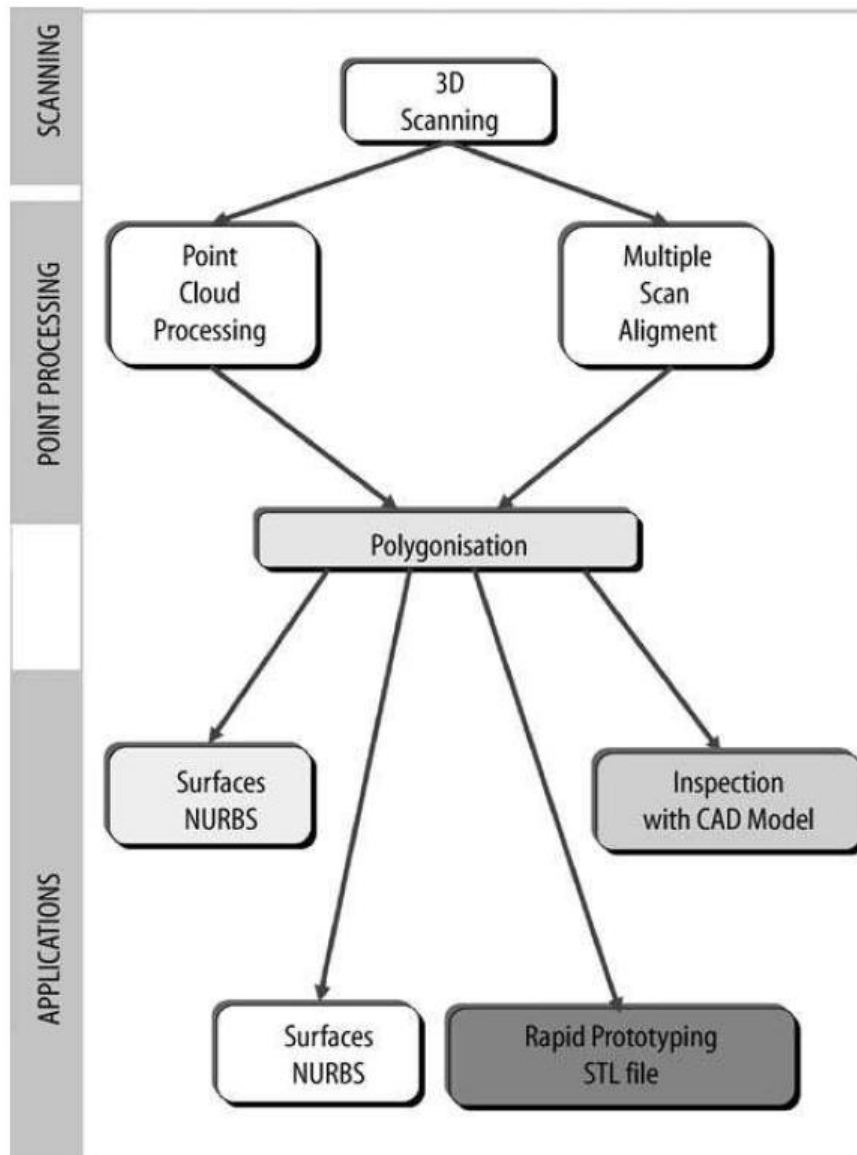
4.2.2 STEPS IN REVERSE ENGINEERING



Steps in Reverse Engineering

1. SCANNING:

- Involves developing a scanning strategy to capture complete part geometry.
- Generally, this step includes:
 - Selecting the appropriate scanning technique.
 - Preparing the part to be scanned.
 - Performing the actual scanning process.
- The goal is to capture all geometric features such as steps, slots, pockets, and holes.
- 3D scanners are used to record the geometry of the part, generating point clouds that define the surface shape.
- There are two main types of scanners:
 1. Contact scanners
 2. Non-contact scanners



Reverse Engineering – Generic Process

2. POINT PROCESSING:

- Involves importing and refining the raw point cloud data obtained from scanning.
- Reduce noise in the scanned data.
- Minimize the number of points without losing essential geometric details.
- Predefined filters are used to clean and optimize the data.
- Multiple scans may be required to capture all geometric features of the part.
- This process often involves rotating the part to scan from different angles.
- Proper planning of multiple scan datums:
 - Reduces time and effort during the point processing phase.
 - Minimize errors during the merging of multiple datasets.

3. APPLICATIONS:

- This phase focuses on creating electronic CAD models from existing physical components using Reverse Engineering techniques.
- After polygonization, the generated 3D model can be used in several applications:

- Surface Model: The polygon model can be converted into smooth mathematical surfaces (NURBS – Non-Uniform Rational B-Splines) for CAD modeling.
- Inspection with CAD Model: The 3D scanned model can be compared with the original CAD design to identify deviations, assess dimensional accuracy, or detect manufacturing errors.
- Rapid Prototyping (STL File): The polygonised model can be exported as an STL file and used for 3D printing, rapid prototyping, or rapid manufacturing.

4.2.3 DIGITIZATION TECHNIQUES

A critical stage in the Reverse Engineering process is digitization, where the physical geometry of a component is captured and converted into digital data. Digitization serves as the bridge between the physical object and its virtual CAD representation. In this phase, the surface geometry of the part is measured using specialized scanning devices that generate dense point clouds or coordinate data.

Digitization techniques play a vital role in determining the accuracy, resolution, and efficiency of the entire reverse engineering workflow. The choice of technique depends on factors such as the size of the object, required precision, surface characteristics, material properties, and the intended application of the final 3D model. In general, digitization methods fall into two main categories: contact scanning and non-contact scanning. Each method has its own strengths, limitations, and suitable areas of application.

The following sections describe the two major digitization techniques used in Reverse Engineering:



Contact scanning



Non-contact scanning

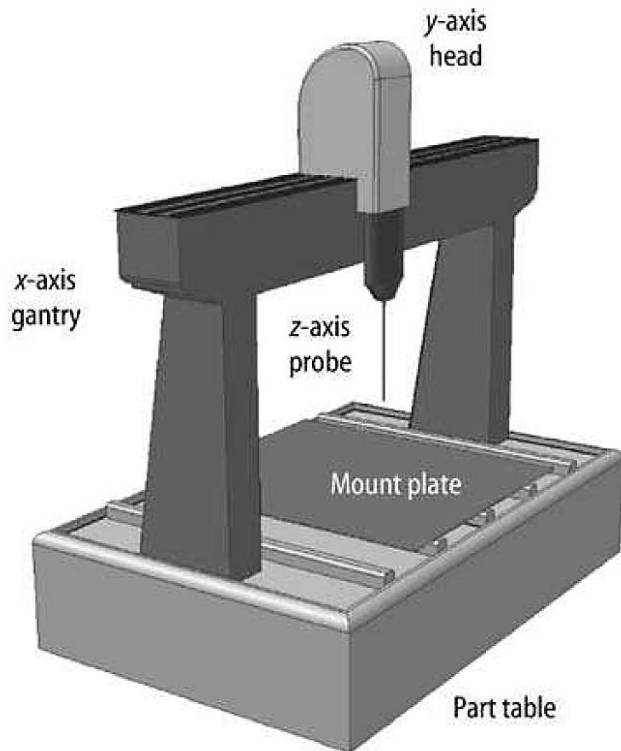
Methods of Scanning a part in Reverse Engineering

1. CONTACT SCANNING:

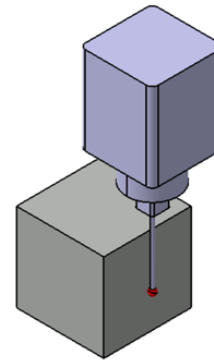
- This technique uses contact probes that automatically trace the contours of a physical surface.
- Conventional CMM is used for smaller objects with intricate detail or a Portable CMM Arm-Based Digitizing (for e.g.: FARO Platinum Arm) is used for small to medium objects with intricate to moderate detail.
- Provide high accuracy, typically within a tolerance range of ± 0.01 to ± 0.02 mm.
- Each data point is generated sequentially at the probe tip.
- The probe deflects slightly to register a point, maintaining constant contact pressure during scanning.

Limitation:

- The process is relatively slow, especially for large or complex parts, since points are collected one at a time.
- The contact pressure makes it unsuitable for soft or flexible materials (e.g., rubber), as it can deform the surface and reduce accuracy.



Coordinate Measuring Machine



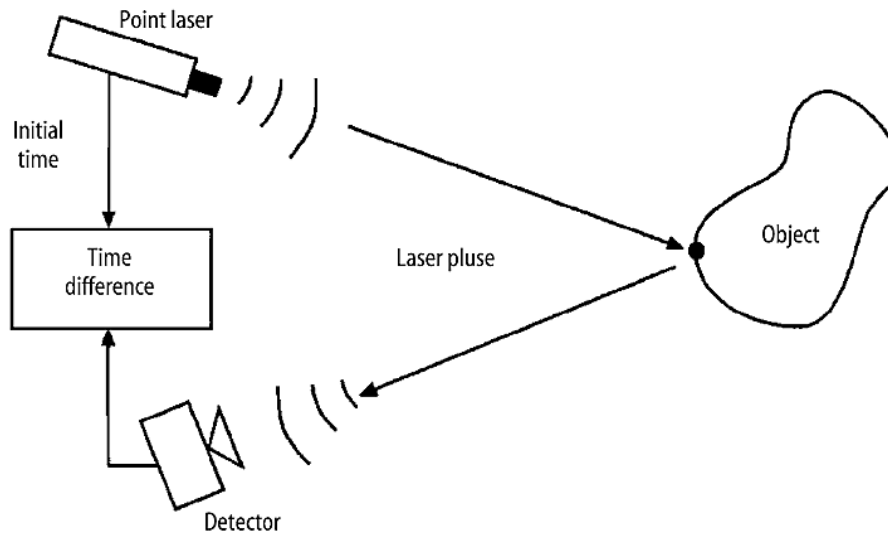
Touch probe

2. NON-CONTACT SCANNING:

- This technique capture data without physical contact with the part.
- Utilize lasers, optics, and charge-coupled device (CCD) sensors to collect point data.
- Capable of capturing large amounts of data quickly compared to contact methods.
- Typical tolerance range: ± 0.025 mm to ± 0.2 mm.
- High data acquisition speed - suitable for scanning complex geometries rapidly.



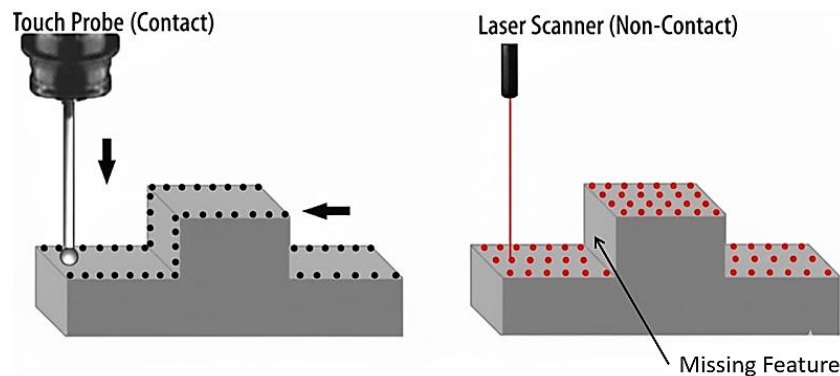
3D Laser Scanner



Principle of non-contact laser scanning

Limitation:

- Generally, it is less accurate than contact methods. Better suited for applications where speed is more important than precision.
- Difficulty in capturing data from surfaces parallel to the laser axis.



- Shiny or reflective surfaces can cause light reflection problems. Such surfaces often require a temporary fine powder coating to reduce reflectivity.

4.2.4 DATA PROCESSING FOR ADDITIVE MANUFACTURING

Although AM systems such as SLA, FDM and SLS differ significantly in how they process materials liquid photopolymers, solid thermoplastic filaments, and polymer powders respectively or in general in any AM technology the core data-processing workflow used to convert a CAD model into a printable format remains fundamentally similar.

All systems rely on:

1. Preparing the CAD model
2. Orienting the part
3. Generating supports
4. Slicing the model into layers
5. Generating the tool path

1. CAD MODEL PREPARATION:

The data processing stage begins with the creation or acquisition of a three-dimensional CAD model that represents the desired component. Once the model is finalized, it must be exported into a neutral file format capable of representing complex geometries using tessellated surfaces. When exporting to STL, the Angle tolerance and chord height must be chosen such that the surface is represented accurately without creating unnecessary data density.

After export, the model undergoes a thorough verification to ensure it is suitable for layer-by-layer fabrication. This involves checking that the model is watertight, free from gaps, overlapping triangles, intersecting surfaces, or inverted normals. Any defects must be repaired using mesh-editing software to prevent slicing errors or incomplete tool paths. The model may then be optimized by removing redundant facets, modifying wall thicknesses, adding fillets, or scaling the design to fit within the build volume of the AM system.

Overall, a clean, error-free, and properly optimized CAD model is essential for achieving accurate and reliable AM builds.

2. PART ORIENTATION:

Part orientation is one of the most influential decisions in AM because it affects surface finish, dimensional accuracy, mechanical performance, support requirements, thermal behavior, and build time. Each part must be oriented relative to the build platform such that the geometry is produced efficiently and with minimal surface defects.

An important consideration during orientation is the staircase effect, a form of surface roughness that occurs on sloped or curved surfaces due to layer discretization. Orienting such surfaces horizontally reduces this effect, while vertical alignment may exaggerate it. Orientation also affects the overall build time, since parts placed vertically tend to increase the number of layers compared to parts placed horizontally.

Mechanical properties are influenced by orientation because many AM processes produce anisotropic structures. Aligning critical load-bearing features along the strong directions of the process can significantly enhance part performance.

Finally, orientation determines how easily trapped regions, cavities, or enclosed voids can be drained, cleaned, or depowdered during post-processing. A well-chosen orientation strikes a balance between surface quality, accuracy, strength, support minimization, and manufacturing efficiency.

3. SUPPORT GENERATION:

AM builds objects layer by layer, and certain geometrical features cannot be fabricated without temporary structural support. Support generation is therefore an essential step in most AM workflows.

Supports are required when a particular region of the part does not have sufficient underlying material in earlier layers to build upon. Examples include overhangs, isolated islands, thin projections, internal bridges, and steep slopes.

Supports perform several critical functions:

- They prevent deformation or collapse caused by gravity, thermal stresses, or process forces.
- They help stabilize tall or thin structures during fabrication.

- They improve heat dissipation, recoating reliability, and layer adhesion in certain processes.
- They ensure proper material deposition or curing by providing a foundation for unsupported regions.

After fabrication, supports are removed through breaking, cutting, melting, dissolving, or depowdering, depending on the material and support type. Effective support design aims to strike a balance between stability, material usage, ease of removal, and minimal impact on surface finish.

4. MODEL SLICING:

Model slicing is the digital process that transforms a 3D model into a series of 2D cross-sections representing individual layers of the final component. These slices define the exact geometry that will be fabricated at each step, making slicing a fundamental bridge between CAD data and machine instructions.

The slicing plane is perpendicular to the build direction, and the chosen layer thickness significantly impacts part quality. Thin layers offer superior surface finish and higher dimensional accuracy but increase build time because more layers are required. Thick layers reduce the number of layers and potentially speed up fabrication but may increase staircase effects, reduce precision, and require higher process energy.

Many modern slicers support adaptive slicing, where layer thickness is varied throughout the build. Sloped or curved surfaces are assigned thinner layers to reduce geometric errors, while flat or less detailed regions use thicker layers to shorten production time.

Although thicker layers may reduce the total number of layers, they do not always decrease build time because material deposition rates, laser scanning speed, curing time, and recoating dynamics are also influenced by layer thickness. Therefore, selecting optimal layer thickness is a balance between accuracy, efficiency, and process stability.

5. TOOL PATH GENERATION:

In this final stage, the sliced model is translated into precise machine instructions that guide the additive manufacturing system during the build. The process involves converting each sliced layer into tool paths such as G-code, scan vectors, or other machine-specific commands, depending on the AM technology used.

Key aspects include controlling parameters such as print speed, material deposition rate or laser power, scan patterns, and temperature settings. Effective tool path planning also incorporates optimization strategies like minimizing non-productive travel moves, reducing retractions, smoothing directional changes, and ensuring uniform heat distribution—especially important in metal AM.

Overall, tool path generation plays a crucial role in determining dimensional accuracy, surface finish, mechanical performance, and the overall reliability of the printed part.